

Software Design Specification (SDS)

Project Title: MindEase

Empowering Lives Through Accessible, Confidential, and Timely Crisis Response

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Document History

Version	Date	Authors	Change Log	Approved By
V1.0	11 Sept 2026	Team 5	Initial Version	

1. Introduction

1.1 Purpose

This Software Design Specification (SDS) describes the architectural, component-level, and data design for **MindEase**, a centralized web-based mental health crisis support platform. The document translates the functional and non-functional requirements defined in the MindEase Software Requirements Specification (SRS) into technical architectural decisions, database schemas, subsystem component structures, API definitions, and deployment specifications.

This SDS serves as the technical blueprint for software developers, system architects, QA engineers, administrators, and evaluators responsible for implementation, testing, and system maintenance.

1.2 Scope

MindEase is a responsive web application designed to connect individuals experiencing emotional distress or crisis situations with trained peer volunteers, professional psychologists, and partner NGOs. The system design encompasses multi-role access control, real-time encrypted messaging, professional video consultation integration, resource management, and administrative moderation tools.

In-Scope Architecture:

- Responsive web application frontend accessible across desktop, tablet, and smartphone browsers.
- Centralized backend architecture for handling crisis support requests, peer chat routing, and risk escalation workflows.
- Professional consultation module with integrated video communication capabilities.
- Referral pipelines connecting users to verified partner NGOs for specialized cases (e.g., domestic violence, legal/social support).
- Volunteer lifecycle management including account verification, shift scheduling, and continuous 24/7 coverage tracking.
- Verified mental health resource hub and payment-integrated mental health workshop module.
- End-to-end data security, encrypted real-time chat, and strict Role-Based Access Control (RBAC).

Out-of-Scope Architecture:

- AI-based automated mental health diagnosis, AI counseling, or automated risk assessment chatbots.
- Automated medical prescriptions, medication recommendations, or clinical treatment plans.
- Native iOS and Android mobile applications (focused strictly on responsive web architecture).

- Direct system integrations with government emergency dispatch systems or hospital EMR databases.

1.3 Intended Audience

- **Developers:** Implementing the React Native app, Node.js/Express APIs, Socket.io chat, and E2EE security modules.
- **System Architects:** Designing WebRTC video integrations, database schemas, and component workflows.
- **QA Engineers:** Crafting test suites for peer triage, therapist bookings, and encrypted mood logs.
- **Clinical Moderators:** Verifying therapist credentials and enforcing platform safety guidelines.
- **Project Managers & Evaluators:** Tracking implementation milestones and assessing academic project compliance.
- **Future Maintainers:** Referencing system architecture, API contracts, and security constraints.

1.4 References

- IEEE Std 1016-2009 – IEEE Standard for Information Technology—Systems Design—Software Design Descriptions.
- MindEase Software Requirements Specification (SRS), Version 1.0 (09 February 2026).
- MindEase Project Proposal, Department of CSE, University of Asia Pacific (Version 1.0).
- IEEE Std 830-1998 – IEEE Recommended Practice for Software Requirements Specifications.
- National Mental Health Survey 2018-19, Bangladesh.
- Web Content Accessibility Guidelines (WCAG) 2.1 & Web Security Standards.

2. System Architecture & Design Overview

2.1 Design Goals

- **Confidentiality & Privacy-First Architecture:** Protect sensitive mental health data through role-based access control, encrypted communication, and explicit user consent protocols before sharing information with external stakeholders.
- **Timely Crisis Escalation & Seamless Pathways:** Enable immediate transition of high-risk cases from peer volunteers to verified psychologists or specialized partner NGOs without data loss.
- **Continuous 24/7 Service Availability:** Maintain continuous coverage through automated volunteer availability tracking and shift scheduling.
- **Low Cognitive Friction & High Accessibility:** Deliver a responsive, lightweight web interface accessible on desktop and mobile browsers, ensuring ease of use during high-distress situations.
- **Ethical Boundary Enforcement:** Enforce strict system boundaries where peer volunteers provide initial support without making medical diagnoses, and automated AI/ML models are explicitly excluded from crisis counseling.
- **Modularity and Extensibility:** Decouple core platform functions—such as real-time messaging, video consultations, workshop payments, and resource management—into modular components.

2.2 Design Principles

- **Separation of Concerns:** Keep client-side web rendering, core application workflows, database persistence, and external service integrations in distinct operational layers.
- **Role-Based Access Control (RBAC):** Restrict system features and data access strictly based on assigned user roles (Support Seeker, Volunteer, Psychologist, NGO Representative, Workshop Facilitator, Administrator).
- **High Cohesion and Low Coupling:** Design modules—such as the Crisis Management Engine, Referral Gateway, and Payment Processor—to interact through clean backend APIs rather than direct state modification.
- **Data Minimization:** Collect and retain only essential user and case details required to deliver support, maintain records, and conduct referrals.
- **Graceful Degradation:** Ensure public-facing features like emergency helplines, suicide hotline directories, and self-help articles remain fully functional during chat engine maintenance or network latency.

2.3 System Context

MindEase operates as a centralized web application that interacts with a cloud-hosted backend server, a relational database, and integrated third-party services. The primary external actors and external services are:

- **Individuals Seeking Support (primary end-users)** who submit crisis support requests, communicate through secure real-time chat, access verified mental health resources, receive professional or NGO referrals, and register for workshops.
- **Trained Volunteers** who manage shift availability, accept active crisis support requests, conduct peer chat triage, and escalate high-risk cases requiring professional or institutional intervention.
- **Professional Psychologists** who review escalated high-risk cases, conduct text or video consultations, guide peer volunteers, and maintain confidential client progress notes.
- **Partner NGOs** who receive specialized case referrals (such as domestic violence, abuse, or legal aid cases) and update assigned case resolution statuses.
- **Workshop Facilitators / Doctors** who create and host mental health awareness workshops, manage session schedules, and upload psychoeducational materials.
- **System Admins / Moderators** who verify volunteer, psychologist, and NGO credentials, manage role-based access control (RBAC), moderate reported content, and maintain the resource hub.
- **External services:** Modern Web Browsers (Chrome, Firefox, Edge, Safari for client-side rendering), Video Communication SDK/API (for psychologist video consultations), and Payment Gateway Services (for processing paid workshop enrollment fees).

The system boundary encloses the web application frontend interfaces, the backend REST APIs, the real-time chat messaging engine, and the centralized relational database storing user profiles, active crisis requests, referral logs, and verified mental health resources. Everything outside this boundary—including external client web browsers, payment processor servers, and third-party video communication infrastructure—is treated as an external entity.

3. System Architecture

3.1 Architectural Overview

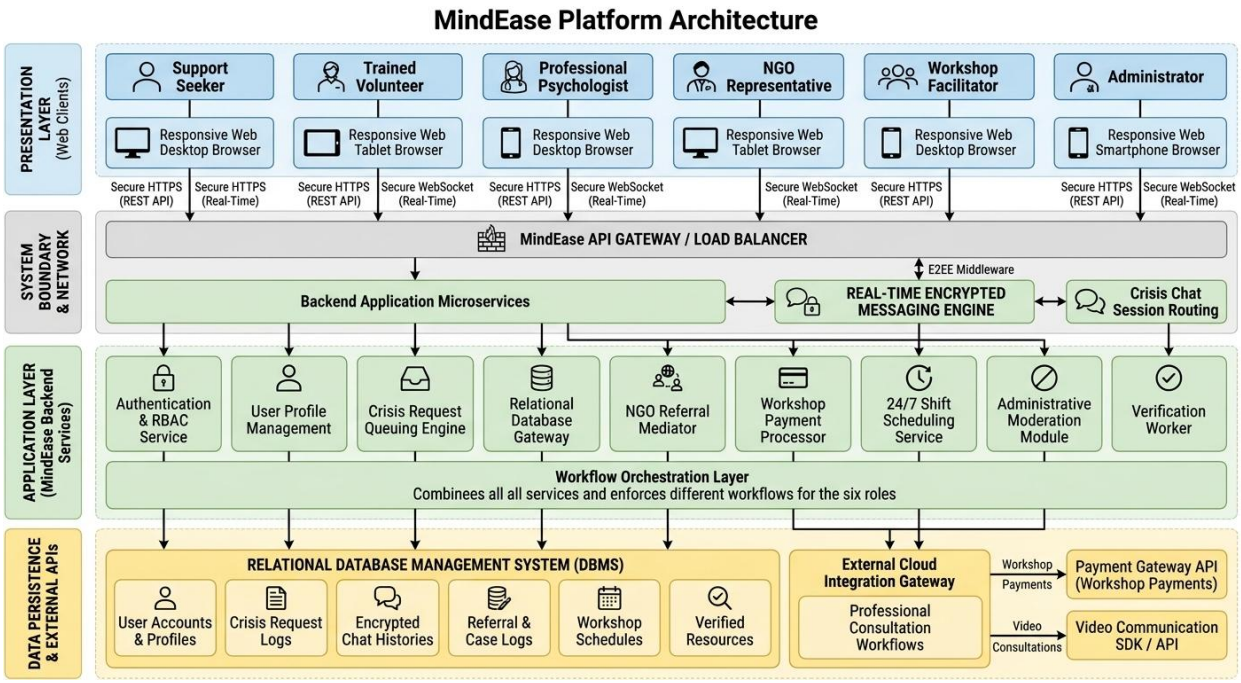
MindEase follows a multi-tier client–server architecture with a strong emphasis on real-time encrypted communication, user confidentiality, and role-based access control. The high-level style is a combination of:

- **Layered Architecture on the web client** (Presentation / UI Component Layer → Client-Side Business Logic → State Management & API Communication Layer).
- **RESTful backend services** that expose secure resources for user authentication, resource hub management, volunteer shift scheduling, workshop processing, and administrative moderation.

- **Event-driven real-time messaging engine** (WebSocket protocol) that manages instant, low-latency text messaging and live session state updates between support seekers, peer volunteers, and verified psychologists during active crisis chats.

The web client is responsible for rendering responsive interfaces across desktop and mobile browsers, managing live chat UI states, validating user inputs, and rendering video consultation frames. The backend is responsible for enforcing Role-Based Access Control (RBAC), managing crisis request queues, routing chat sessions to available volunteers, executing risk escalation workflows to psychologists or partner NGOs, and processing workshop payment transactions. Relational database management system (RDBMS) queries and foreign key indexes are utilized so that user records, referral logs, shift schedules, and verified mental health resources remain transactional, consistent, and secure

3.2 Architectural Diagram



3.3 Technology Stack

The MindEase platform is built on the MERN stack (MongoDB, Express.js, React.js, Node.js) to deliver secure, real-time crisis support across desktop and mobile web browsers.

Layer	Technology	Purpose
Frontend	**React.js**	Responsive single-page application for desktop and mobile web browsers.
Backend	**Node.js + Express.js**	Event-driven RESTful API server executing core business logic.
Database	**MongoDB (Atlas)**	Document-based NoSQL storage for user profiles, chat logs, and resources.
Real-Time Engine	**Socket.io**	Low-latency, end-to-end encrypted WebSocket engine for crisis chats.
Video Gateway	**WebRTC / Agora SDK**	Web-native video streaming framework for professional consultations.
Auth & Security	**Firebase Auth + JWT**	Phone OTP authentication paired with Role-Based Access Control (RBAC).
Payment Gateway	**SSLCommerz / Stripe**	Secure payment processing gateway for mental health workshop fees.
Hosting & CI/CD	**Render & Vercel**	Cloud-native hosting pipelines with automated continuous deployment.

3.4 Deployment Architecture

The MindEase platform is built on a fully cloud-native infrastructure, eliminating physical on-premise servers and native app store distribution requirements.

- **Frontend Hosting:** React.js single-page application hosted on Vercel/Netlify and distributed via global Content Delivery Networks (CDNs) for fast loading on desktop and mobile web browsers.
- **Backend Services:** Node.js, Express RESTful APIs, and the Socket.io WebSocket engine hosted on Render, running behind mandatory HTTPS/WSS protocols for secure TLS network transport.
- **Database Instance:** Fully managed MongoDB Atlas cloud database connected to the Render backend service via encrypted connection strings with daily automated backups.
- **Third-Party Integrations:** Browser and server-level integration gateways for Firebase Authentication (Phone OTP), WebRTC/Agora video SDKs, and SSLCommerz/Stripe payment processors.
- **Scalability & Maintenance:** Designed for horizontal auto-scaling, allowing additional Node.js backend instances to run behind cloud load balancers as concurrent user traffic grows.

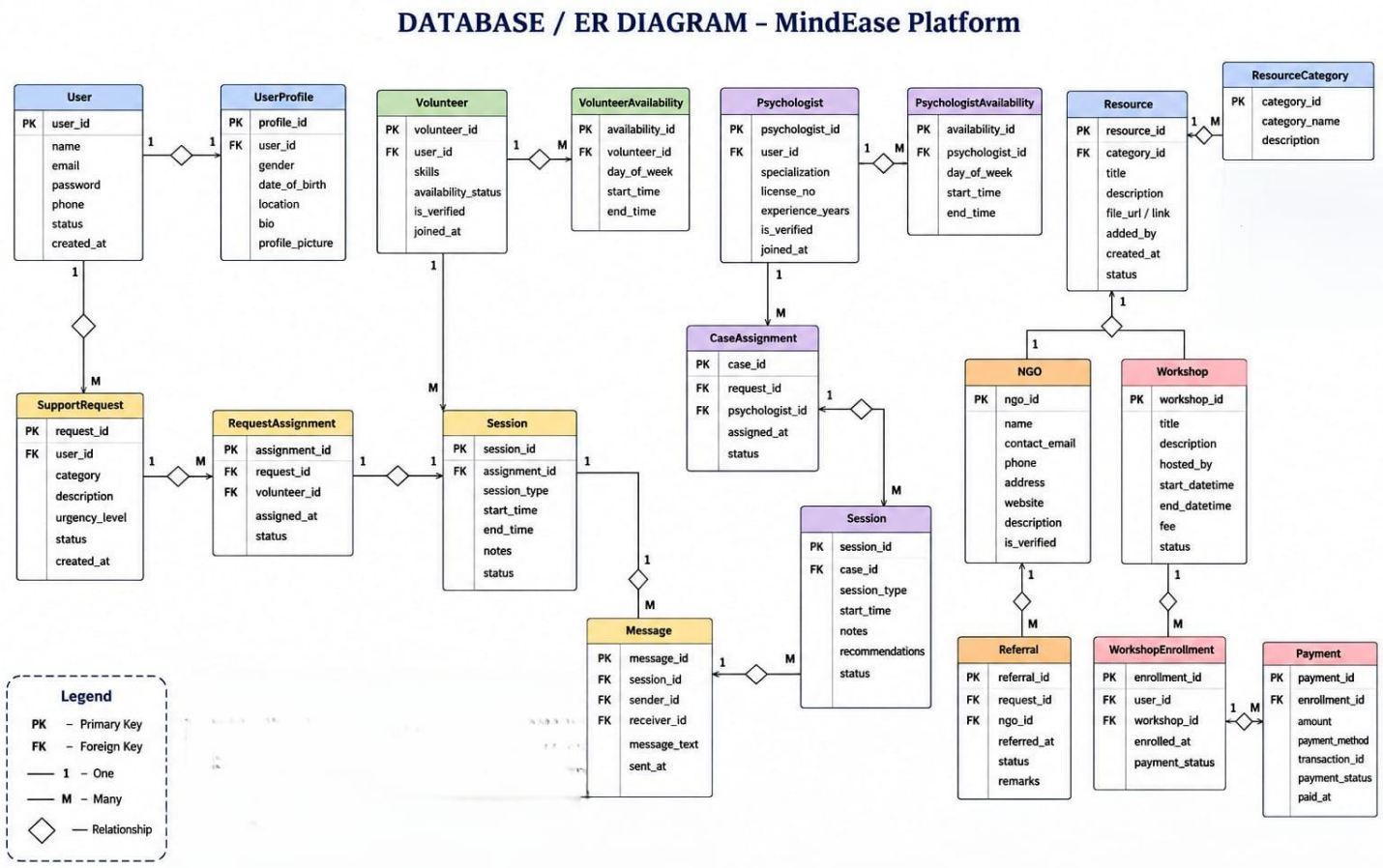
4. Data Design

4.1 Data Model Overview

MindEase utilizes a flexible, document-oriented NoSQL data model powered by MongoDB Atlas. This model is optimized for low-latency web interactions, real-time WebSocket messaging, and dynamic schema growth across multi-role workflows (Support Seekers, Volunteers, Psychologists, NGOs, and Admins).

- **Document-Based Architecture:** Core data entities are structured as JSON/BSON collections (e.g., `Users`, `CrisisRequests`, `ChatSessions`, `Referrals`), minimizing complex relational joins and providing fast read/write operations for active crisis sessions.
- **Client-Side Browser Storage:** The React.js frontend uses browser-native `LocalStorage` and `IndexedDB` to handle active session caching, JWT tokens, and temporary chat UI states. Mobile native storage mechanisms (such as SQLite or AsyncStorage) are not used.
- **Privacy & Encrypted Records:** Sensitive fields—including user journal entries, risk self-assessments, progress summaries, and E2EE chat payloads—are encrypted before being written to the central database store.
- **Directory Indexing:** Location queries are strictly limited to static geospatial coordinates of verified partner NGOs, crisis centers, and hospitals for directory filtering. Continuous user location tracking, travel routes, and device movement histories are explicitly excluded.

4.2 Entity Relationship Diagram (ERD)



4.3 Data Dictionary

The following dictionary details the core attributes, data types, system usage, and constraints across all primary entities defined in the MindEase database schema.

Entity	Key Attributes	Data Type	Description	Constraints
User	`user_id`, `email`, `phone`, `status`	UUID, VARCHAR, ENUM	Core account details and login state	PK, Unique, Not null

UserProfile	profile_id, user_id, bio	UUID, TEXT	Secondary demographic and bio profile	PK, FK (User.user_id)
Volunteer	volunteer_id, availability_status, is_verified	UUID, ENUM, BOOL	Peer supporter credentials and readiness	PK, FK (User.user_id)
Psychologist	psychologist_id, license_no, experience_years	UUID, VARCHAR, INT	Clinical professional credentials	PK, FK (User.user_id), Unique
SupportRequest	request_id, urgency_level, status	UUID, ENUM	Submitted distress ticket and severity	PK, FK (User.user_id), Default 'pending'
Case / Request Assignment	assignment_id, request_id, volunteer_id / psychologist_id	UUID	Assignment link for peer or psychologist	PK, FK (SupportRequest, Volunteer/Psychologist)
Session	session_id, session_type, status	UUID, ENUM	Peer or clinical support session details	PK, FK (Assignment/Case)
Message	message_id, session_id, message_text	UUID, TEXT	E2EE encrypted chat payload	PK, FK (Session.session_id)
NGO & Referral	referral_id, request_id, ngo_id, status	UUID, ENUM	Institutional referral for partner NGOs	PK, FK (SupportRequest, NGO.ngo_id)
Workshop & Payment	payment_id, transaction_id, payment_status	UUID, VARCHAR, ENUM	Workshop enrollment financial record	PK, Unique transaction code

4.4 Data Storage & Retention

Central storage retains only hashed authentication credentials, verified psychologist and volunteer profiles, partner NGO directories, workshop event schedules, and moderation audit logs. All stored chat message payloads remain end-to-end encrypted (E2EE), ensuring no unencrypted clinical session notes or private journal entries are readable on central servers. Support requests and interaction logs older than 18 months are aggregated into anonymized usage metrics, and the raw session logs are purged, subject to academic and legal review.

On-device web storage manages temporary session tokens (JWT) and client-side encryption keys using secure browser storage (`IndexedDB` / encrypted `LocalStorage`). Queued support requests and encrypted message payloads are permanently deleted from client memory immediately following successful transmission.

5. Component Design

5.1 Component Overview

Component Name	Description	Responsibility
Auth & RBAC Module	User registration, authentication, and role management	Password hashing, JWT token issuance, RBAC enforcement across Seekers, Volunteers, Psychologists, and Admins.
Crisis Triage & Request Engine	Distress request processing and automated dispatch	Evaluating urgency levels, pairing Seekers with available Peer Volunteers, or escalating to Psychologists.
Real-Time E2EE Chat Engine	Low-latency encrypted messaging service	Socket.io WebSocket handling, client-side encryption key management, message delivery acknowledgments.
Clinical Care & Booking Gateway	Professional consultation scheduling	Psychologist slot management, session link generation, clinical case history tracking.

Institutional Referral Gateway	Outbound support connection to external partners	Transferring high-risk or specialized cases to verified partner NGOs and crisis centers.
Workshop & Resource Portal	Educational hub and community event management	Publishing psychoeducational materials, managing workshop listings, and processing registrations.
Payment Engine	Transaction processing for paid services	Integration with SSLCommerz/Stripe gateways, payment verification, and audit logging.
Moderation & Safety Dashboard	Administrative oversight and platform compliance	Reviewing reported interactions, managing profile verification, and recording audit logs.

5.2 Detailed Component Design

Component: Real-Time E2EE Chat Engine

- **Purpose:** Provide secure, low-latency, end-to-end encrypted messaging between Seekers and assigned Supporters (Volunteers or Psychologists) without storing plaintext on central servers.
- **Inputs:** User message text, active `session_id`, client-side secret key, WebSocket connection trigger.
- **Outputs:** Client-side encrypted ciphertext payload, real-time message broadcast, database write acknowledgment.
- **Dependencies:** Web Crypto API, Socket.io client/server library, Node.js JWT verifier, MongoDB (`Session` and `Message` collections).
- **Internal Logic:** User sends message → encrypt text client-side via Web Crypto API → emit socket event → server validates sender JWT → route payload to active socket room → persist ciphertext in `Message` collection → emit delivery receipt back to sender.
- **Sequence (simplified):** Client UI → Web Crypto Module → Socket Client → Node.js Socket Server → MongoDB (`Message` collection) → Recipient Socket Client → Decryption → Recipient UI.

Component: Crisis Triage & Support Request System

- **Purpose:** Process incoming distress tickets, determine severity, and immediately route the Seeker to appropriate peer or professional care.

- **Inputs:** Support category, free-text description, self-reported distress level, Seeker `user_id`.
- **Outputs:** Active `SupportRequest` record, automated `RequestAssignment` (to Peer Volunteer) or `CaseAssignment` (to Psychologist).
- **Dependencies:** Triage rules engine, `VolunteerAvailability` service, `PsychologistAvailability` service, Notification module.
- **Internal Logic:** Seeker submits ticket → analyze category and urgency level (`low`, `medium`, `high`, `critical`) → if `low`/`medium`, match against active Volunteers (`availability_status = available`) → create `RequestAssignment` → if `critical`, assign directly to an available Psychologist or trigger an immediate NGO Referral → alert assigned supporter via push notification.

Component: Clinical Care & Booking Gateway

- **Purpose:** Manage professional therapy appointments, securely handling scheduling, payment, and video session access.
- **Inputs:** Target `psychologist_id`, selected `availability_id`, payment gateway token.
- **Outputs:** Confirmed `Session` record, WebRTC video room token, `Payment` audit record.
- **Dependencies:** Payment Engine (SSLCommerz/Stripe SDK), WebRTC/Agora video gateway, `PsychologistAvailability` collection.
- **Internal Logic:** Seeker selects available schedule slot → lock slot temporarily → route user to Payment Engine → upon gateway confirmation callback, set status to `paid` → instantiate `Session` record with unique WebRTC video room link → update `PsychologistAvailability` status → dispatch calendar invitations to both parties.

6. Interface Design

6.1 User Interface Design

The MindEase user interface is built with a calming, empathetic, and highly accessible aesthetic designed to reduce cognitive load during emotional distress. It prioritizes rapid navigation and psychological safety, allowing users to quickly request support, enter encrypted chat sessions, or access self-help tools. Major views include Dashboard/Home, Crisis Triage, Peer & Clinical Chat, Therapist Booking, Mood Journaling, Resource & Workshop Hub, and Profile/Settings.

6.1.1 Wireframes

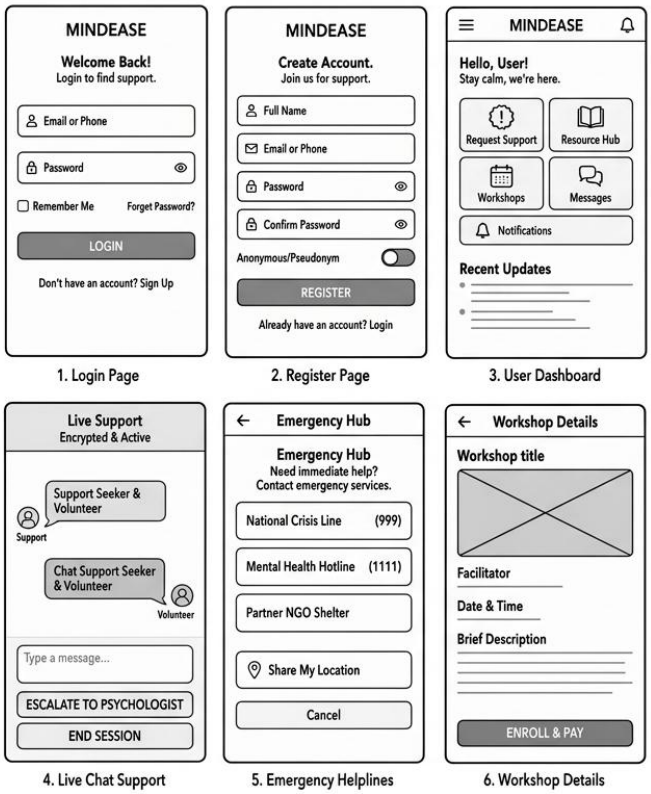
The wireframes utilize a responsive, web-first design system optimized for mobile viewports (390 × 844 px) and desktop displays (1440 × 900 px). The Dashboard provides single-tap access to immediate crisis assistance and emergency support channels, while dedicated secondary screens support end-to-end encrypted messaging, professional consultation scheduling, anonymous reflection journaling, and community workshops.

The primary web navigation structure is defined as: **Dashboard → Crisis Support → Chat & Consultations → Resources & Workshops → Profile & Settings**

6.1 USER INTERFACE DESIGN

A. WIREFRAMES

The following wireframes show the basic layout of major screens in the MindEase system.



B. UI STANDARDS

The following standards will be followed throughout the MindEase system.

UI Element	Standard
Design Style	Calm, inclusive, and minimal
Color Palette	Primary: #2E5BFF (Blue) Secondary: #5E35B1 (Purple) Accent: #00897B (Teal/Green) Danger: #E53935 (Red) Background: #F5F7FA (Light Gray)
Typography	Font Family: Poppins / Inter Base Font Size: 14px-16px
Buttons	Rounded corners (8px), clear labels, consistent colors
Icons	Simple, consistent line icons
Navigation	Bottom Navigation (Mobile), Sidebar (Web)
Forms	Clear labels, placeholder text, real-time validation
Feedback	Success, error, and info messages are clearly shown
Spacing	Consistent spacing using 4px grid system
Responsiveness	Mobile-first design, responsive for all devices

C. ACCESSIBILITY CONSIDERATIONS

The following accessibility guidelines will be followed to ensure the system is usable by all users.

☑ Use sufficient color contrast between text and background (WCAG AA standard).
☑ Use readable font sizes and avoid overly decorative fonts.
☑ Do not rely only on color to convey important information. Use text or icons as well.
☑ Provide clear labels and instructions for all form fields.
☑ Ensure all buttons and interactive elements are large enough and easily clickable.
☑ Provide alternative text for images and icons where necessary.
☑ Ensure compatibility with screen readers and keyboard navigation.
☑ Provide clear error messages and triggers to help users correct mistakes.
☑ Keep the interface simple and consistent to reduce cognitive load.

6.1.2 UI Standards

MindEase maintains a clean, modern, and empathetic visual style across all screens to foster psychological safety and minimize cognitive friction during distress. MindEase Blue (`#1A90F8`) serves as the primary accent color, paired with a soft light background (`#F8F9FA`) and neutral secondary tones (`#6C757D`) for complementary elements. Accessible typography (Inter/Poppins), 8px rounded UI components, standardized 4px grid spacing, and clear, minimal line icons are applied systematically across the interface.

The bottom navigation bar on mobile viewports and persistent sidebar on desktop maintain consistent route states, with active views visually highlighted to ensure seamless navigation across all primary mental health modules.

6.1.3 Accessibility Considerations

The MindEase interface prioritizes high contrast (WCAG AA compliant), clear typography, and minimal cognitive load to ensure the platform remains effortless to navigate during periods of emotional distress. Core action points—such as initiating peer support, booking a counseling slot, or viewing helpline directories—are designed with prominent, touch-friendly tap targets (minimum 48×48 px) and intuitive iconography.

Future technical implementations will support complete screen-reader integration (ARIA attributes), dynamic text scaling, alternative text descriptions for all graphic elements, and a sensory-friendly dark mode to accommodate users with visual impairments or neurodivergent needs.

6.2 External Interfaces

- **Payment Gateways (SSLCommerz / Stripe):** Secure transaction processing for therapist appointment bookings and community workshop enrollments.
- **Telehealth & Video Gateway (WebRTC / Agora SDK):** Encrypted, low-latency audio and video streaming infrastructure for live clinical consultation sessions.
- **Authentication Service (Twilio / Custom OTP & JWT):** Manages phone/email OTP verification, token generation, and secure role-based session control.
- **Real-Time WebSockets (Socket.io Engine):** Transmits low-latency, end-to-end encrypted (E2EE) chat payloads between Seekers, Peer Volunteers, and Psychologists.
- **Cloud Storage (AWS S3 / Cloudinary):** Encrypted storage for static psychoeducational content, downloadable self-care resources, and user profile media.

6.3 API Design

Endpoint	Method	Request Payload	Response Payload	Description
`/auth/otp/request`	POST	`{ phone }`	`{ status, message }`	Sends OTP to user's registered phone number
`/auth/otp/verify`	POST	`{ phone, otp }`	`{ token, user }`	Verifies OTP and returns JWT auth token
`/triage/request`	POST	`{ urgency_level, category, notes }`	`{ ticket_id, status, assigned_supporter }`	Submits distress ticket for peer/psychologist matching
`/chat/messages`	POST	`{ session_id, encrypted_payload }`	`{ message_id, timestamp, status }`	Persists E2EE ciphertext message payload
`/therapists`	GET	`?specialization=&availability=`	`[{ therapist_id, name, slots }]`	Retrieves list of verified psychologists and slots

/bookings/create	POST	{ therapist_id, slot_id, payment_method }	{ booking_id, payment_url, status }	Initiates consultation booking & payment redirect
/journal/entries	POST	{ mood_score, reflection, tags }	{ entry_id, created_at }	Saves encrypted personal mood and wellness log
/workshops	GET	?category=&page=	[{ workshop_id, title, date }]	Fetches available workshops and self-help tools

6.4 Hardware Interfaces

MindEase runs entirely on commercial off-the-shelf mobile and desktop devices without custom hardware dependencies:

- **Audio & Video Peripherals:** Microphone and camera hardware utilized for WebRTC-based clinical teleconsultations.
- **Biometric Sensors:** Native fingerprint scanners and facial recognition hardware (Touch ID / Face ID / Android Biometrics) for client-side local session lock and privacy protection.
- **Display & Touch Inputs:** Standard capacitive screens and desktop input devices for interface navigation and journal entry recording.

6.5 Communication Interfaces

- **Protocols:**
  - **HTTPS (TLS 1.2+):** Encrypted RESTful transport for user authentication, booking processing, and resource management.
  - **WSS (WebSocket Secure):** Persistent, low-latency socket channel for real-time E2EE messaging between Seekers and Supporters.
  - **WebRTC / SRTP:** Secure Real-Time Transport Protocol for encrypted end-to-end video and audio therapy sessions.
- **Message Formats:**
  - **REST / Sockets:** Standard JSON payloads for API commands and state transfers.
  - **Encrypted Payload:** Symmetric key ciphertext strings (`AES-GCM-256`) for chat transmission and database persistence.

7. Security Design

7.1 Authentication & Authorization

MindEase utilizes phone or email One-Time Password (OTP) verification for secure user onboarding and login. Successful validation issues a short-lived JSON Web Token (JWT) along with a secure refresh token. Returning user sessions support native biometric unlock (Touch ID / Face ID / Android Biometrics) for quick local access without bypassing server-side security.

Role-Based Access Control (RBAC) is enforced across four distinct user roles:

- **Seeker:** Accesses mood journaling, requests peer support, books therapist sessions, and enrolls in wellness workshops.
- **Volunteer:** Receives peer-triage requests and conducts anonymous support chats.
- **Psychologist:** Manages session slots, conducts WebRTC video consultations, and reviews clinical histories.
- **Admin / Moderator:** Verifies professional therapy credentials, moderates flagged resources, and oversees system operations.

7.2 Data Security

- **At Rest on Device:** Mood logs, local settings, and encryption keys are stored using platform-native secure storage (iOS Keychain / Android KeyStore).
- **In Transit:** Standard REST traffic is secured over TLS 1.3, chat payloads run over Secure WebSockets (WSS), and video streams utilize SRTP.
- **End-to-End Encryption (E2EE):** Peer and counseling chat messages are encrypted client-side using `AES-GCM-256`. Plaintext message content is never accessible to central backend servers.
- **Anonymity & Privacy:** Seekers can operate using customizable pseudonyms. Direct identifiers (such as phone numbers) are stripped from active chat metadata.
- **Data Retention:** Automated backend lifecycle jobs permanently purge unreferenced session logs and temporary media buffers after 18 months.

7.3 Security Threat Analysis

Threat	Impact	Mitigation
Unauthorized Access via Stolen Device	Exposure of private journal entries or active chat logs	Native biometric app-lock requirement, session auto-timeout, and remote token revocation.
Identity Exposure & Stigma	Exposure of mental health history or therapy records	Zero-knowledge server architecture, pseudonymous chat handles, and client-side E2EE.
Unverified / Fraudulent Therapists	Ineffective or unsafe guidance provided to users	Mandatory manual admin verification of clinical licenses and credentials prior to profile activation.

Man-in-the-Middle (MitM) Interception	Interception of real-time therapy messages or auth tokens	Strict TLS 1.3 enforcement, WebSocket SSL wrapping, and SSL certificate pinning on mobile apps.
Central Database Breach	Leakage of account details or user metadata	Column-level PII encryption, hash-based token verification, and zero central storage of plaintext messages.

8. Performance Design

8.1 Performance Requirements

- **Peer Triage & Matching:** Peer supporter or psychologist matching requests processed and assigned within 3 seconds under normal network conditions.
- **Real-time Messaging Latency:** End-to-end encrypted (E2EE) chat payload transmission completed within 500 ms over WebSocket (WSS) connections.
- **Video Call Signaling:** WebRTC peer connection setup and audio/video handshake completed within 2 seconds.
- **Concurrent Capacity:** System maintains response times and session integrity for at least 5,000 concurrent active users across messaging, booking, and journaling channels.

8.2 Scalability Strategy

- **Horizontal API & WebSocket Scaling:** Node.js Express backend and Socket.io instances scale horizontally behind an elastic load balancer, using a Redis Pub/Sub adapter to sync multi-node chat connections.
- **Database Optimization:** Optimized database indexing (e.g., compound indexes on `user_id`, `therapist_id`, `created_at`, and booking status) ensures fast query execution for journal entries and consultation schedules.
- **Caching Layer:** Redis cache holds active therapist slot availability, system configuration, and public wellness resources with short TTL.
- **Client-side Processing:** End-to-end encryption/decryption execution is offloaded to client devices, minimizing backend CPU overhead.
- **Media Delivery:** Video streams are peer-to-peer via WebRTC (routed through TURN/STUN relay servers when NAT traversal is required), and static educational content is delivered via cloud storage and CDN.

8.3 Capacity Planning

For the academic prototype and early public pilot, a single modest cloud instance (2 vCPU, 4 GB RAM) and a managed database instance will handle initial load. Growth beyond a few thousand daily active users will trigger horizontal API instances, Redis cluster nodes, and database read

replicas. Mobile client battery and data usage are optimized through event-driven WebSocket connections and lazy-loaded assets rather than aggressive polling.

9. Error Handling & Logging

9.1 Error Handling Strategy

All network operations, database queries, and encryption/decryption routines are wrapped in robust error-handling structures. User-facing error messages remain clear, empathetic, and non-technical (e.g., *"Connection lost – attempting to reconnect your session..."*). Local mood journal entries write to native secure storage first and sync asynchronously to the server via exponential backoff retries when connectivity is restored.

9.2 Logging Strategy

- **Client-Side:** Minimal diagnostic logs (devoid of PII, mood content, or plaintext messages) stored locally in debug mode for developer diagnostics.
- **Server-Side:** Structured JSON logging capturing request IDs, trace IDs, log levels (`DEBUG`, `INFO`, `WARN`, `ERROR`), and administrative audit trails.
- **Privacy Controls:** Encrypted chat payloads and confidential journal text are explicitly excluded from application logs to preserve user confidentiality.

10. Deployment Design

The mobile front-end is built using standard React Native CLI / Expo toolchains and signed for Android/iOS distribution. The Node.js/Express REST and WebSocket backend is containerized (Docker) and hosted on cloud platforms (e.g., Render/AWS). Automated GitHub Actions pipelines handle testing and continuous deployment.

10.1 Deployment Strategy

- **CI/CD Pipeline:** Linting → Unit & Integration Tests → Docker Image Build → Container Deployment → Signed Mobile Build Generation.
- **Versioning:** Semantic versioning (`MAJOR.MINOR.PATCH`) aligned with app store release build codes.

10.2 Environment Configuration

Environment	Purpose	Configuration
Dev	Local development & debugging	Local emulator, mock OTP verification, mock payment sandbox, local debug backend URL.

QA	Integration & acceptance testing	Staging server, test phone numbers, sandbox credentials for SSLCommerz/Stripe payment testing.
Prod	Live user & clinical operations	Production cloud cluster, real SMS gateway, live SSLCommerz/Stripe keys, app store production builds.

10.3 Rollback Strategy

Backend deployments are version-tagged; previous container images can be redeployed within minutes via automated rollback hooks. Mobile clients support forced-update notifications for critical security patches, while offline local capabilities (such as writing local journal entries) remain accessible even during temporary server maintenance.

11. Design Constraints

- **Regulatory & Privacy:** Must strictly comply with Bangladesh personal data protection principles and health data privacy standards (data minimization, explicit consent, zero surveillance). Sensitive mental health interactions and journal records require strict confidentiality and client-side encryption.
- **Technology:** Primary target platform is Android API 24+ (Android 7.0 or higher), optimized for low-to-mid-range smartphones prevalent in Bangladesh. Client-side cryptographic operations (E2EE) must maintain low CPU and battery consumption.
- **Budget:** Reliance on open-source libraries, free-tier cloud platforms (e.g., Render, MongoDB Atlas, Redis Cloud), and open-source WebRTC protocols for the academic prototype phase.
- **Timeline:** Core functional modules (OTP Auth, Pseudonymous Peer Triage, E2EE Chat, Mood Journaling, and Therapist Slot Booking) must be fully demonstrable within a single-semester academic schedule.
- **Connectivity:** Design must account for low-bandwidth or intermittent cellular data in rural/suburban regions, allowing users to write mood journal entries offline and auto-sync when network connectivity returns.

12. Assumptions & Dependencies

Assumptions

- Users possess a compatible Android smartphone and will grant necessary network, audio/video (for consultations), and biometric permissions.
- Seekers prefer pseudonymous peer interactions to avoid social stigma associated with mental health support.

- Trained peer volunteers and verified licensed psychologists are available to handle triage requests and scheduled therapy sessions during the pilot.
- Payment gateway sandbox environments (SSLCommerz / Stripe) remain accessible for integration testing of therapy bookings.
- Initial psychoeducational resources, self-help tools, and mental health guides can be curated from accredited clinical sources.

External Dependencies

- **Android SDK:** Local biometric authentication, storage APIs, and media device access (Camera/Microphone).
- **WebRTC / Agora SDK:** Infrastructure for encrypted real-time video/audio teleconsultations.
- **Cloud Hosting & Database:** Managed backend hosting (Render) and database clusters (MongoDB Atlas, Redis).
- **OTP Service:** SMS gateway or authentication service (Twilio / Firebase Auth) for phone number verification.
- **Payment Gateways:** SSLCommerz and Stripe APIs for transaction processing and booking confirmation.
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13. Traceability Matrix

The following matrix links high-priority functional and non-functional requirements from the MindEase System Requirements Specification (SRS) to the design components that realize them:

Requirement ID	Design Component	Remarks
FR-1.1 – FR-1.5	Authentication & RBAC Module	Phone/Email OTP validation, pseudonymous user handles, and role-based access control across four user types.
FR-2.1 – FR-2.5	Peer Triage & Matching Engine	Algorithmic distress classification, automated peer volunteer assignment, and crisis escalation protocols.
FR-3.1 – FR-3.4	E2EE Real-Time Messaging Engine	Client-side `AES-GCM-256` encryption, WebSocket Secure (WSS) transmission, and zero server-side plaintext storage.

FR-4.1 – FR-4.4	Professional Therapist Booking Gateway	Psychologist slot scheduling, SSLCommerz/Stripe payment processing, and WebRTC video signaling integration.
FR-5.1 – FR-5.4	Wellness & Mood Journaling Module	Local mood tracking, reflection logs, encrypted data persistence, and offline sync capability.
FR-6.1 – FR-6.3	Workshop & Resource Directory	Curated psychoeducational content library, self-help interactive tools, and community workshop enrollment.
FR-7.1 – FR-7.4	Admin & Credential Verification Portal	Clinical license verification workflow, flagged chat moderation, role management, and system audit logging.
NFR-1.1 – NFR-1.5	Security & Privacy Architecture	Client-side E2EE, PII metadata stripping, TLS 1.3 network transport, and native biometric session protection.
NFR-2.1 – NFR-2.3	Offline Sync & High Availability Engine	Local-first encrypted storage for journal entries, exponential backoff sync, and automatic Socket connection recovery.

14. Appendices

Appendix A – Glossary

- **MindEase:** Digital platform for peer support, encrypted counseling, wellness tracking, and mental health teleconsultation.
- **E2EE:** End-to-End Encryption (`AES-GCM-256`) ensuring plaintext chat messages are accessible only to session participants.
- **WebRTC:** Web Real-Time Communication protocol used for high-quality, encrypted video/audio clinical consultations.
- **RBAC:** Role-Based Access Control (Seeker, Peer Volunteer, Licensed Psychologist, Admin).
- **JWT:** JSON Web Token used for secure, stateless session authentication.
- **OTP:** One-Time Password for phone/email account verification.

- **Triage Engine:** Algorithmic routing system that classifies emotional distress level and assigns incoming Seekers to Peer Volunteers or Psychologists.
- **FR / NFR:** Functional Requirement / Non-Functional Requirement.

Appendix B – Diagrams

The following architectural and data flow diagrams are maintained in the project repository to provide a comprehensive representation of MindEase across functional, database, and structural perspectives:

- High-level system architecture diagram
- Entity-Relationship Diagram (full cardinality)
- Data Flow Diagrams (Context Level & Level 1)
- Use Case Diagram

1. Data Flow Diagram (DFD)

The Data Flow Diagram illustrates how information flows across the MindEase platform between its primary stakeholders: Seekers, Peer Volunteers, Licensed Psychologists, and Admins/Moderators.

Seekers interact with the platform to submit distress triage requests, participate in pseudonymous E2EE chats, book therapy sessions, record encrypted mood logs, and register for wellness workshops. Peer Volunteers accept assigned triage requests to provide peer counseling. Licensed Psychologists set up consultation availability, review clinical case histories, and conduct teleconsultations. Admins verify professional credentials, oversee resource moderation, and audit platform security logs.

2. Database Requirement Diagram

Identifies the core data entities required to support MindEase operations. The central entities include:

- **Users & Roles:** Seeker, Peer Volunteer, Psychologist, Admin.
- **Support & Clinical Entities:** Triage Ticket, E2EE Chat Session, Encrypted Message, Psychologist Profile, Slot Availability, Appointment Booking, and Payment Transaction.
- **Self-Care & Content Entities:** Journal Entry, Mood Metric, Workshop, Resource Category, and System Audit Log.

3. Use Case Diagram

Defines system interactions categorized by primary actors:

- **Seeker:** Register/login, manage pseudonymous profile, complete distress triage, chat with peer supporters, book and attend video therapy sessions, write encrypted mood logs, and access self-help resources.

- **Peer Volunteer:** Set availability, receive triage queue assignments, conduct anonymous E2EE chat sessions, and escalate high-risk cases to clinical support.
- **Psychologist:** Manage availability slots, view client appointment logs, conduct WebRTC video/audio therapy sessions, and update clinical notes.
- **Admin / Moderator:** Validate clinical licenses/credentials, moderate public workshop listings, review flagged interactions, and monitor system diagnostics.

4. Entity Relationship (ER) Diagram

Presents the logical schema structure centered around the base **User** entity. Specialized attributes extend into Seeker, Volunteer, and Psychologist profiles. Key relational mappings link:

- Users to Triage Requests and E2EE Chat Sessions.
- Seekers to encrypted Journal Entries and Mood Logs.
- Psychologists to Availability Slots, Bookings, and Payment Records.
- Administrative accounts to Credential Verification Logs and Moderation Records.

5. High-Level Architecture Diagram

Illustrates the end-to-end technical infrastructure of MindEase:

- **Client Layer:** React Native mobile application and web portal for Seekers, Therapists, and Admins.
- **API & Service Layer:** Node.js / Express backend processing RESTful endpoints and Socket.io engine handling low-latency WebSocket communication.
- **Core Micro-Services:** Authentication Service, Triage Engine, E2EE Chat Dispatcher, Booking Gateway, and Journal Sync Module.
- **Data & Storage Layer:** MongoDB Atlas for primary persistent records, Redis for active session caching and socket pub/sub, and AWS S3 for static self-help media.
- **External Integrations:** SSLCommerz / Stripe for payment processing, WebRTC/TURN infrastructure for video streaming, and SMS/OTP gateways.

Drive Link:

https://drive.google.com/drive/folders/14khgSmp79a8kO5sb0UifWxedIKW8E610?usp=drive_link

Appendix C – Supporting Documents

- MindEase Project Proposal (v1.0)
- MindEase System Requirements Specification (SRS v1.0)
- MindEase Figma Interactive Wireframe & Component Library
- OpenAPI / Swagger API Specification (v1.0)